

SURAJ RAJ

Portfolio [LinkedIn](#) [GitHub](#) [LeetCode](#)

Email: rajsuraj181002@gmail.com

Mobile: +91 9661508657

EDUCATION

- Indian Institute of Technology, Mandi** Himachal Pradesh, India
B.Tech, Major in Civil Engineering; CGPA: 7.67/10 Nov 2021 – Jun 2025

EXPERIENCE

- Wall Climber Bot - IIT Mandi iHub and HCI Foundation** Mandi, HP
Embedded Systems Developer Jun 2023 – Aug 2023
 - Developed real-time control software on Arduino to independently coordinate 8 motors across 4 arms, implementing resource-constrained embedded logic to manage suction timing and multi-directional movement simultaneously.
 - Iteratively redesigned the control algorithm across 3 hardware versions based on testing outcomes, reducing mechanical complexity and improving climbing efficiency by 30% on vertical surfaces.

SKILLS

- Languages:** Python, C++, SQL, HTML, CSS, JavaScript
- Libraries and Frameworks:** FastAPI, REST API, Flask, OpenCV, NumPy, Pandas, Scikit-Learn, SQLAlchemy, Face Recognition
- Databases and Tools:** PostgreSQL, MySQL, SQLite, Git, GitHub, VS Code, Docker

PROJECTS

- Payment Orchestration Layer** [Github Link](#)
Python, FastAPI, PostgreSQL, Docker, pytest, pydantic Jun 2026
 - Built a backend payment routing system across 4 gateways (Razorpay, Stripe, PayU, UPI) with automatic failover, circuit breaker pattern, and multi-criteria scoring algorithm to get an optimal gateway per transaction.
 - Implemented idempotency service preventing duplicate charges, HMAC-SHA256 webhook signature verification, and a 23-state transaction state machine enforcing valid payment lifecycle transitions.
 - Achieved 84 automated tests across 13 failure scenarios with full Docker Compose deployment; system starts from scratch with a single command.
- AI-Powered SQL Chatbot** [Website Link](#) [Github Link](#)
Python, Flask, Gemini, MySQL, SQLAlchemy, HTML, CSS, JavaScript Oct 2024
 - Developed an AI-driven NLP application using LangChain, Google Gemini 1.5 and Flask to dynamically generate and execute SQL queries based on user input and database schema, providing a contextual response.
 - Built a secure, conversational user interface with HTML, CSS and JavaScript, enabling real-time interaction to translate natural language queries into actionable SQL results.
- Real-time Face Recognition Attendance System** [Website Link](#) [Github Link](#)
Python, Flask, OpenCV, face recognition, MySQL, HTML, CSS, JavaScript, REST API Jun 2024
 - Created a full stack, real-time attendance system using Flask and OpenCV leveraging dlib's face recognition (99.38% accuracy) to identify individuals from live video feeds, automating attendance with high reliability.
 - Engineered a robust backend with Flask-SQLAlchemy and a SQL database, replacing manual record-keeping with a scalable solution for secure student and attendance data management.
 - Implemented an authenticated Admin Dashboard to manage student registrations, view attendance records and export data to Excel.

RELEVANT COURSES

Data Structures and Algorithms	Object-Oriented Programming	Data Science
Database Management Systems	Operating System	Computer Networks

POSITIONS OF RESPONSIBILITY & ACHIEVEMENTS

- Coordinator - Artgeeks club. The official position in student gymkhana for the head of the club.
- Decoration Head - Exodia, the annual cultural festival of IIT Mandi.
- Core Member - Nirmaan club and Artgeeks Club.
- Solved 500+ problems on LeetCode and 200+ on GeeksforGeeks (Data Structures & Algorithms).
- GATE CS 2026 Rank 1000.